

Exploiting k-mers for genotype-phenotype association in spinach

Lukas Schönmann, Heinz Himmelbauer, Juliane C. Dohm

Institute of Computational Biology, Department of Biotechnology, Universität für Bodenkultur (BOKU), Vienna, Austria

Contact: lukas.schoenmann@boku.ac.at



Abstract

Genome-wide association studies (GWAS) are widely employed to find associations between traits and genetic variants. Traditionally, single-nucleotide polymorphisms (SNPs) are used as input. While there have been many successful applications of GWAS, they are also limited in explaining complex traits. Novel approaches using short subsequences (*k*-mers) as input were developed to overcome some of these limitations but come with their own challenges. In the present work, we compared SNP-based and *k*-mer-based GWAS to identify candidate regions that may regulate agronomical traits in spinach, a leafy green plant with high agricultural importance. Illumina whole-genome sequencing data of 303 cultivated and wild spinach accessions were used for the reference-free extraction of *k*-mers. SNP-based GWAS was performed using genetic variants called from the same raw reads along the spinach reference genome. Processing these two different types of input data revealed strengths and weaknesses of each approach. As a complementary approach, we used the *k*-mers as input for an advanced machine learning algorithm and trained random forest models. The models achieved high accuracies for the prediction of downy mildew resistance in spinach. The efficient downstream analysis of *k*-mers (e.g. matching them to relevant genetic loci) is still a challenge but saving the time-consuming steps of genome-wide read mapping and variant calling is advantageous. We show to what extent the different approaches confirmed each other and describe methodological issues that were encountered.

Overview

In this work, we analysed Illumina re-sequencing data of 303 cultivated and wild spinach accessions [1] regarding 20 important agronomical traits related to plant morphology, bolting & flowering behavior, disease resistance and nutritional quality.

Short subsequences of a specific length *k*, so-called *k*-mers, were generated from the raw reads, and the information of their presence or absence in each accession was used as input for a GWAS. Genetic variants from the same raw reads [2] were used to perform a SNP-based GWAS. Both approaches were compared and complemented by random forest machine learning models based on the *k*-mers.

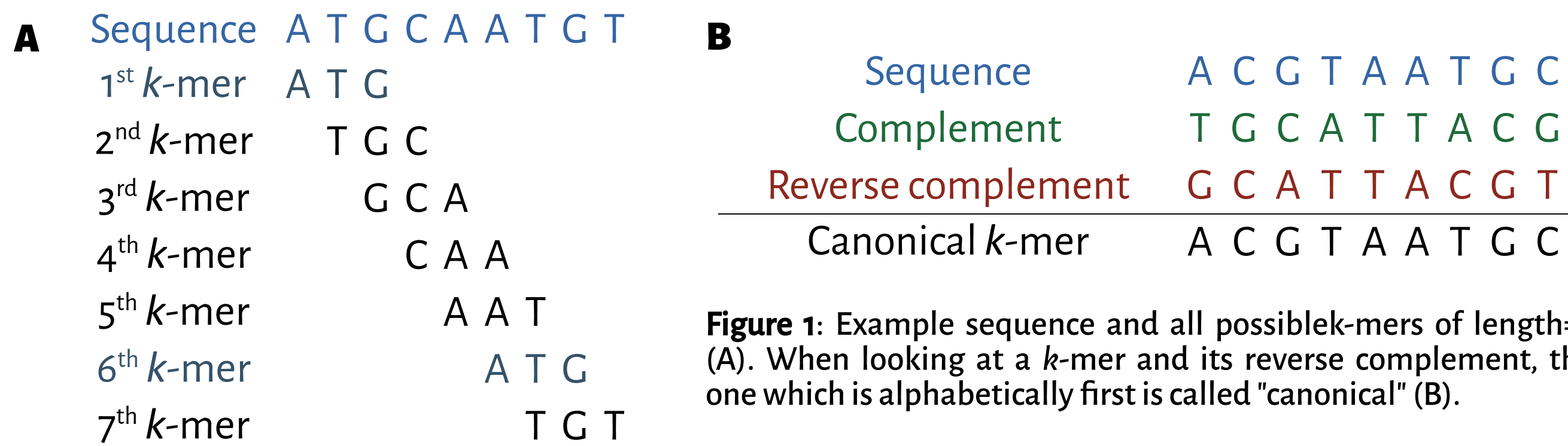
Table 1: Number of sequenced *Spinacia* (S.) accessions per species.

Species	Sequenced accessions
<i>Spinacia oleracea</i> (cultivated)	292
<i>Spinacia turkestanica</i> (wild)	7
<i>Spinacia tetrandra</i> (wild)	3
Unknown species	1
Sum	303

K-mer counting and filtering

A biological sequence represented as string of characters can be split up into shorter subsequences of length *k* (Figure 1 A). Since it is unknown which strand of the double stranded DNA has been sequenced, we only count the canonical *k*-mer (Figure 1 B).

We used a *k*-mer length of 31 and applied the "kmersGWAS" pipeline [2] for counting, filtering and performing *k*-mer based GWAS. Canonical *k*-mers occurring only once were removed, the remaining ones were kept as forward and reverse *k*-mers and were filtered based on two criteria: presence in at least 5 accessions and in both forms in 20 % of accessions. The number of *k*-mers fulfilling these criteria was 858,232,023.



Comparison with a SNP-based GWAS

Table 2: Spinach traits with at least one Bonferroni-corrected SNP at significance threshold $\alpha = 0.05$.

Trait	Number of significant SNPs
Leaf number	223
DM resistance	158
Plant type	92
Petiole color	63
Bolting days	37
Leaf surface texture	9
Petiole length	8
Leaf base shape	5
Leaf width	3
Sex type	2
Leaf apex shape	2
Petiole width	1
Flowering days	1
DM incidence	1

- Variants from a previous study [3] were used to perform a SNP-based GWAS
- 20 traits, 4,580,829 variants processed using Plink 1.9
- After QC, 3,610,120 variants for 303 accessions remained
- Relatedness matrix and association tests calculated with GEMMA
- Significance level $\alpha = 0.05$ was corrected with the Bonferroni method
- Manhattan plots were compared with Cai et al (2021)
- Generally, spikes of significant SNPs can be found in the same regions (Figure 2)
- For key traits like downey mildew (DM) resistance, our SNP-based GWAS also supports the findings of the *k*-mer based approach

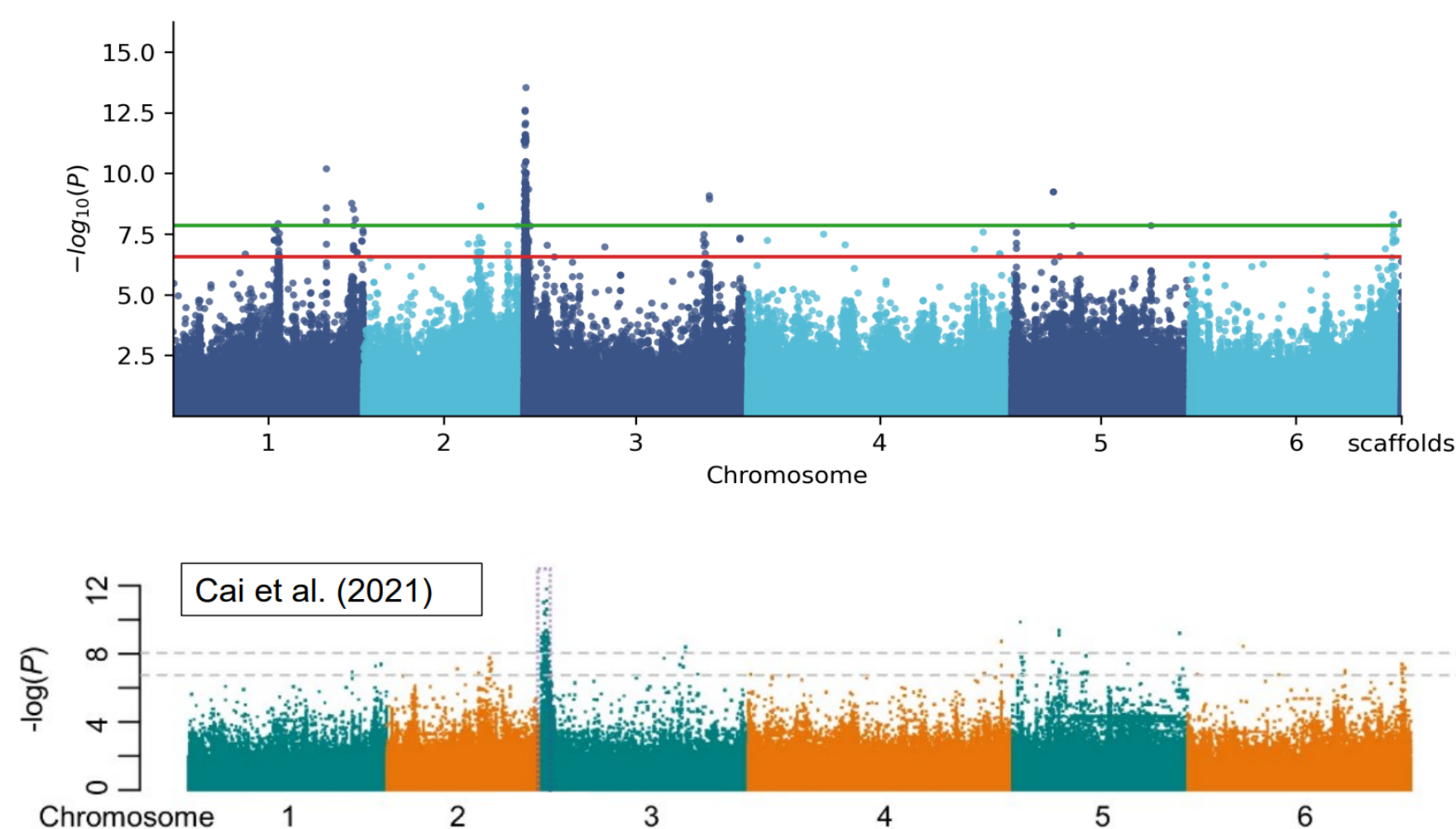
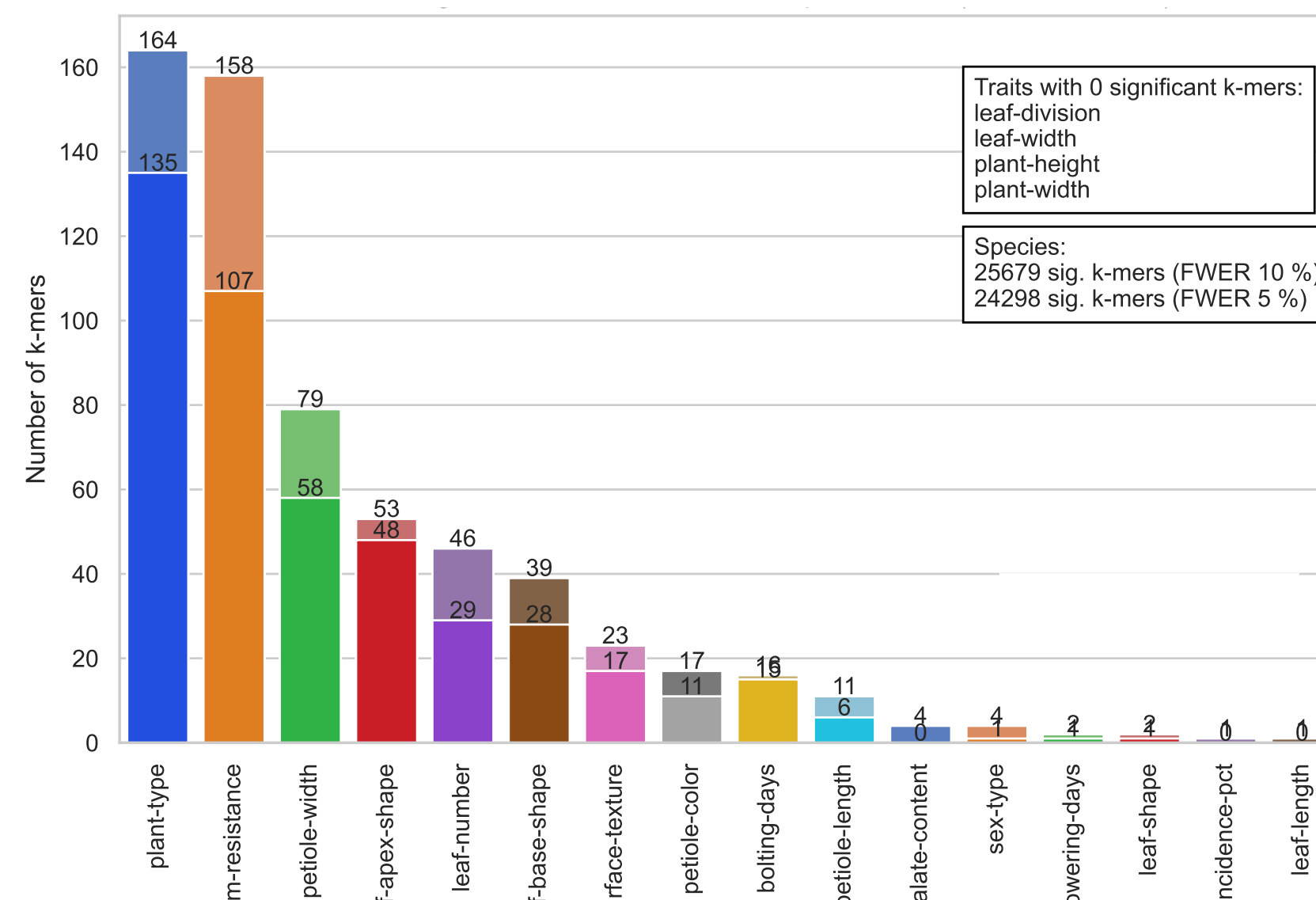


Figure 2: Comparison of Manhattan plots from SNP-based GWAS for the DM resistance trait. Top: present study; bottom: Cai et al. (2021). The spikes at the start of chromosome 3 are located at a disease resistance gene. Horizontal lines indicate the suggestive and genome wide significance thresholds.

Performing a *k*-mer based GWAS

A fast approximate method for association testing (GRAMMAR-Gamma) was used to estimate the variant effects. Only the top-scoring 100,000 *k*-mers were kept for exact association testing running GEMMA. Phenotype permutation was used to calculate *p*-value threshold for 10 % and 5 % family-wise error rate (FWER) significance levels resulting in different numbers of significant *k*-mers for each trait (Figure 3). A specialized tool was developed to efficiently extract sequence reads harboring significant *k*-mers. Those reads were then mapped to the spinach reference to find hits inside genic regions.



A clearly significant region identified for the downy mildew resistance trait was located near gene SOV3g001250 (Figure 4), which is annotated as a putative disease resistance gene and has been identified in a previous study [1].

Figure 3: Number of significant *k*-mers identified for each trait at 5 % (bold bars) and 10 % FWER (muted bars).

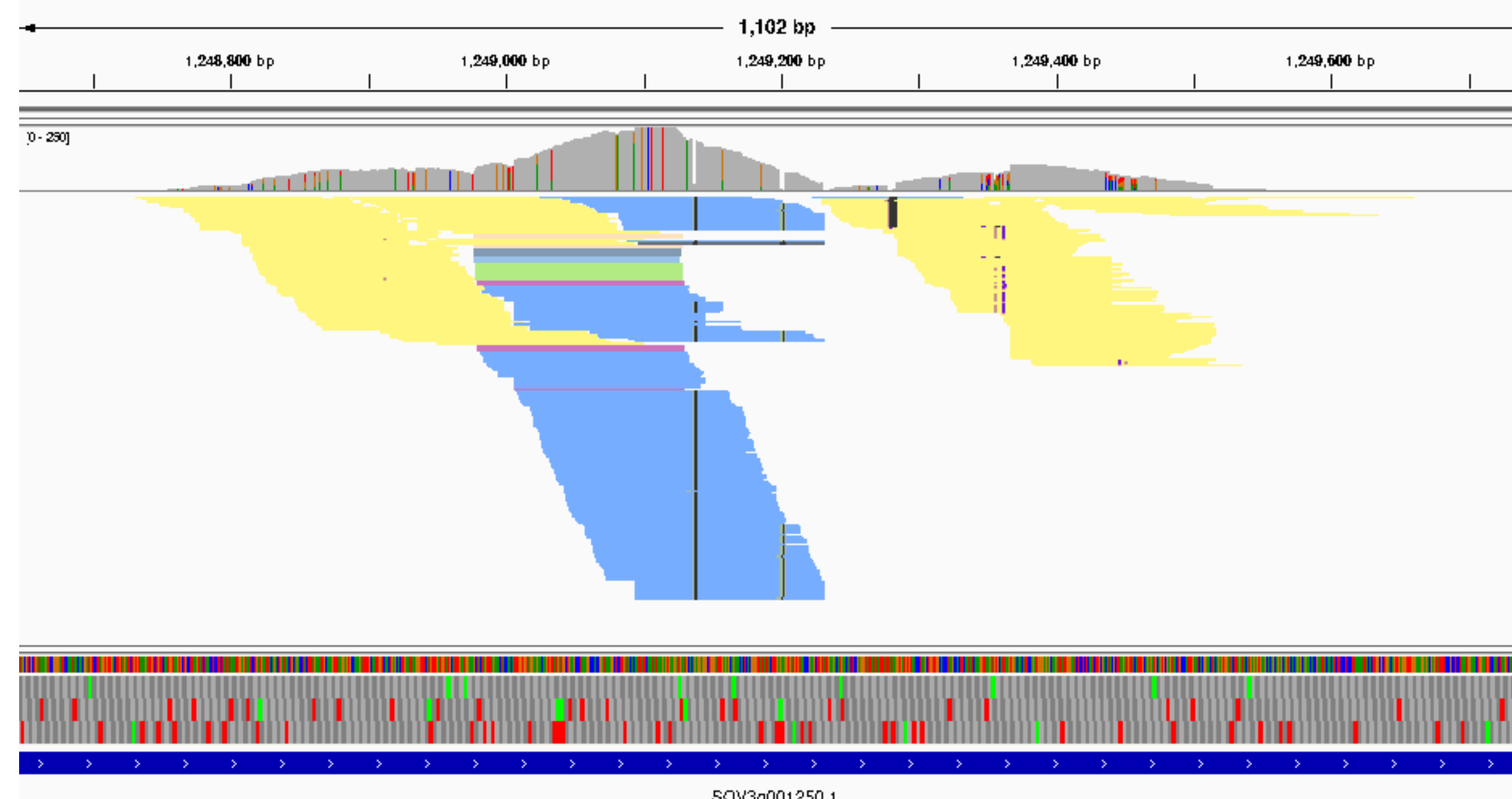


Figure 4: Reads containing *k*-mers with significant association found for downy mildew (DM) resistance. Blue: The *k*-mer is present in this read. Yellow: The mate of the *k*-mer-containing read

Random forest machine learning approach

As a complementary approach, we demonstrate that presence/absence information of *k*-mers in accessions can be used as input for machine learning algorithms. Random forest models were trained on the filtered and reduced datasets of 100,000 *k*-mers. As a proof of concept, we looked only at the DM resistance trait.

A principal component analysis (PCA) computed on this data already hints at the resistance status of the accessions (Figure 5). Before training the models, phenotypic data was coded from five classes to a binary trait for 258 resistant and 37 susceptible accessions. Grid search and cross-validation were utilized to avoid overfitting.

Results

Dummy model (most frequent)	88 %
Random forest (two classes)	96 %
Random forest (without ambiguous accessions)	98 %

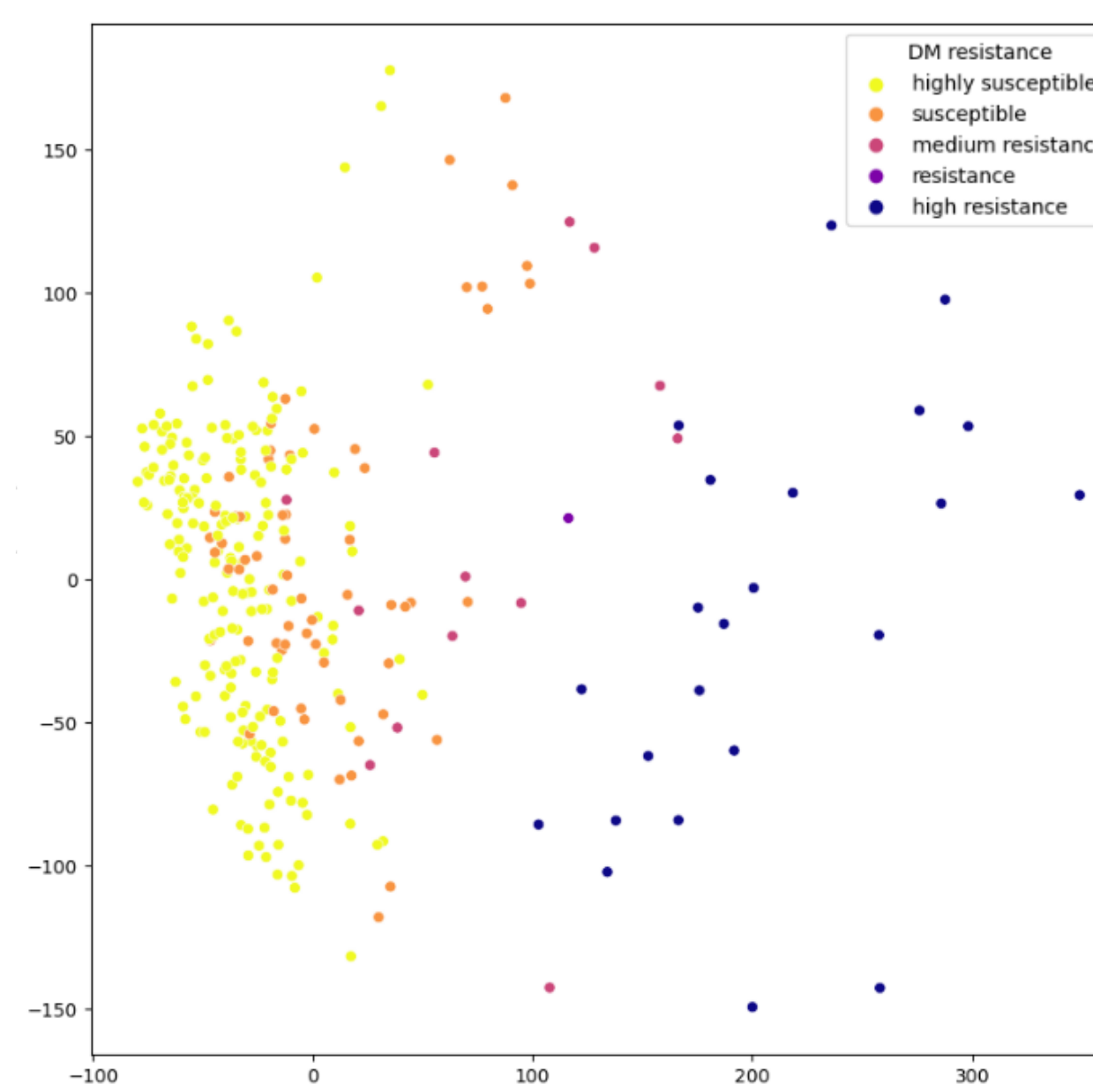


Figure 5: Principal component analysis (PCA) based on presence/absence information of 100,000 *k*-mers from 295 accessions and related to five downy mildew resistance classes.

Summary and conclusion

- *K*-mers were used to effectively find associations between genotype and phenotype without relying on a reference
- *K*-mer-based results were confirmed by a SNP-based GWAS
- By avoiding the time-intensive steps of mapping reads and variant calling, a *k*-mer based GWAS could potentially save time compared to the SNP-based approach
- *K*-mer-based random forest models predicted the disease resistance status of samples with high accuracy
- Downstream analysis of significant *k*-mers leaves room for improvement, e. g. quantitatively expressing the importance of identified regions or visualizing the results

References

- [1] Cai, X et al. Nat Commun 12, 7246 (2021).
- [2] Voichok, Y., Weigel, D. Nat Genet 52, 534–540 (2020).
- [3] Nguyen-Hoang, A. et al. NARGAB 6, 2 (2024).

Acknowledgments

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